Name: Surabhi P
Id: COMETFWC032
Date: 28 August 2025

`https://github.com/surabhi84ard/Mini/tree/8e9042b13304f4b4923d330711d1eb424065bccb/avr-gcc`

Abstract

This manual explains state machines by deconstructing a decade counter.

1 The Decade Counter

The block diagram of a decade counter (which repeatedly counts from 0 to 9) includes:

- Incrementing Decoder
- Display Decoder
- Delay Block (Sequential Logic)

2 Finite State Machine

FSM Structure

Fig. 1 shows the FSM diagram for the decade counter. The state S_0 corresponds to input 0000. The state transition table uses:

- Present state: W, X, Y, Z
- Next state: A, B, C, D

FSM Implementation

The FSM is implemented using D Flip-Flops. The flip-flops hold the input for a duration determined by the clock signal, forming the Delay block.

Hardware Cost

The number of D Flip-Flops required is:

$$\text{No. of Flip-Flops} = \lceil \log_2(\text{No. of States}) \rceil = \lceil \log_2(9) \rceil = 4$$

3 Design Exercises

1. Draw the state transition diagram for a decade down counter (counts from 9 to 0).
2. Write the state transition table for the down counter.
3. Derive the state transition equations with and without don't cares.
4. Verify your design using an Arduino.
5. Design a circuit that detects three consecutive 1s in a bitstream.

References

[1] G. V. V. Sharma. Karnaugh Map. Available online: `https://github.com/gadepall/arduino/raw/master/ide/kmap/gvv_kmap.pdf`